

OCAP: Online Collaborative Adapter Personalization for Personalized Mental-Health Monitoring

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Abstract

Mental-health monitoring uses behavioral data from smartphones or wearable devices to infer users' mental-health states and support timely intervention. As these data arrive continuously and their links to mental-health states may change, online learning is better suited to timely personalization than conventional fixed offline methods. However, personalized mental-health monitoring faces a **low-evidence personalization dilemma**: new users lack enough personal data for reliable personalization, while using other users' data may introduce harmful bias. To address this dilemma, we propose Online Collaborative Adapter Personalization (OCAP) for personalized mental-health monitoring. OCAP uses a shared backbone and one personalized adapter for each user. The backbone learns common associations between sensed behaviors and mental-health states, while each adapter learns the associations specific to its user. Inspired by collaborative filtering, we seek similar users to support personalization when personal evidence is limited. To find such users, we propose a gradient-based grouping method that uses their accumulated model-update directions. These directions directly reflect whether users require compatible model corrections, providing a more reliable grouping signal than raw behavioral similarity. We further propose group refinement, which reassigns a user when another group adapter better explains the user's recent data and prevents fixed sharing relationships from becoming outdated. Finally, extensive experiments on two real-world datasets verify the effectiveness of OCAP.

CCS Concepts

• **Applied computing** → **Health informatics**; • **Computing methodologies** → *Online learning settings*.

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Keywords

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1 Introduction

Recently, advanced smartphones and wearable devices have provided a low-burden way to continuously capture users' behaviors, including sleep, phone interaction, and daily activity. These behavioral data enable personalized mental-health monitoring, which infers users' mental-health states to support timely detection and intervention. A key requirement in this task is personalization because the same sensed behavior can indicate different mental-health states for different users. To this end, existing studies develop the per-user hierarchical predictor I-HOPE [6], build personalized classifiers through collaborative filtering [38], and add individual random effects to random-forest models [20].

However, these personalization methods follow an offline learning paradigm. Their models are trained on fixed historical data and remain unchanged after deployment. This training paradigm fails to handle the streaming characteristic of sensor data, struggling to adapt to evolving behavioral patterns and mental-health states, and leading to degraded performance over time. As shown in Figure 1(a), a model is trained under the offline learning paradigm on the first four temporal windows of the College Experience Study (CES) dataset [25] and then frozen for prediction in later windows. The clear performance drop shows that **associations between sensed behaviors and mental-health states change over time, so a model trained once and kept fixed can become outdated**. Although offline models can be retrained with recent data, the cost of training and deployment delays adaptation to behavioral changes and may reduce the reliability of timely monitoring.

Instead, **online learning enables timely adaptation through a predict-then-update process**. It predicts with the current model and updates the model whenever the corresponding label becomes available. This continual adaptation allows the model to track evolving associations between sensed behaviors and mental-health states without waiting for periodic retraining and redeployment, making

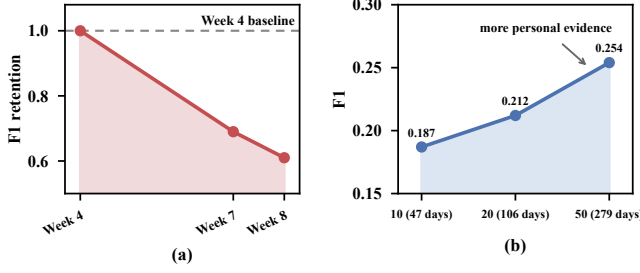


Figure 1: Performance of offline and online models on the CES dataset. (a) F1 retention of a model trained through Week 4 and evaluated on the same users in later weeks. (b) F1 of personalized online models updated only with each user's own data after the first 10, 20, and 50 daily observations, collected over a median of 47, 106, and 279 days, respectively.

it well suited to streaming sensor data. However, directly applying online learning to this task is non-trivial. Figure 1(b) considers a personalized online setting in which each user's model is updated only with that user's own data. F1 is low when few personal observations are available and improves as more observations arrive, but collecting them can take months. Consequently, new users may have only limited personal evidence for a long time. This raises the key question: *How can reliable personalized mental-health monitoring be achieved when personal evidence is limited?*

An intuitive solution is to follow collaborative filtering [15] by identifying similar users and using their information to support personalization. However, direct user matching is unreliable in this streaming setting for two reasons. (1) Similar sensed behaviors do not necessarily imply compatible behavior-state relationships. For example, low mobility may indicate elevated depressive symptoms for a normally active user, but it may simply reflect the usual routine of a user who is consistently less mobile. (2) As user data arrive asynchronously, the users who are suitable for learning together may change over time. Together, limited personal data and unreliable user sharing create the **low-evidence personalization dilemma**: learning from personal data alone is slow, while sharing with incompatible users can introduce harmful bias.

To address these challenges, we propose Online Collaborative Adapter Personalization (OCAP) for personalized mental-health monitoring. OCAP adopts a backbone-adapter predictive architecture consisting of a shared backbone and one lightweight user-specific adapter for each user. The shared backbone learns common patterns shared across users, while each user's adapter captures user-specific associations between sensed behaviors and mental-health states. For each incoming data, the model combines the shared backbone with the corresponding user's adapter for prediction. The initial warm-up data update both components; the backbone is then frozen while the adapters continue learning from subsequent data. To accurately identify compatible users, we propose **gradient-based grouping** strategy, which represents each user through accumulated model-update directions and groups users with similar learning needs. OCAP then averages the member adapters within each group for prediction, allowing limited personal data to be supplemented by data from compatible users. To

prevent these assignments from becoming outdated, **user reassignment** moves a user when another group adapter better explains the user's recent data. Finally, extensive experiments on two real-world datasets demonstrate the effectiveness of OCAP. In particular, OCAP reaches 0.7060 F1 on CES, improving by 0.0522 over the strongest online baseline at 0.6538, and achieves the best GLOBEM F1 at 0.7359.

In summary, our contributions include:

- We identify the **low-evidence personalization dilemma** as a key challenge in online personalized mental-health monitoring, where learning from personal data alone is slow and incompatible sharing can introduce harmful bias.
- We propose OCAP, which addresses this dilemma through gradient-based user grouping, adapter sharing, and user reassignment. To the best of our knowledge, it is the first online learning framework for personalized mental-health monitoring.
- We conduct extensive experiments on two real-world datasets to demonstrate the effectiveness of OCAP.

2 Related Work

2.1 Mental-Health Monitoring

Passive sensing uses smartphones and wearable devices to collect behavioral signals with little user effort, providing a practical foundation for continuous mental-health monitoring [23, 33]. Activity, sleep, mobility, phone interaction, speech, and surrounding context have been studied as behavioral markers of stress, depression, anxiety, and well-being [30, 41]. Longitudinal studies further show that these signals can characterize changes in mental health over weeks, months, and years [25, 34]. Recent work also combines passive sensing with interpretable machine learning to identify which behavioral patterns contribute to individual risk predictions [6].

Existing approaches range from population models that learn common patterns across users to personalized models that incorporate user-specific histories. Personalization can better capture how an individual's sensed behaviors relate to their mental-health state, but it must operate with sparse, noisy, and multimodal observations [24]. Models trained in one population or period also generalize poorly to other users and studies [1, 2, 10].

Although sensing data arrive continuously, most mental-health monitoring approaches remain offline. They train a model on a fixed historical dataset and keep its parameters unchanged during deployment. This static training paradigm assumes that associations between sensed behaviors and mental-health states learned from past data remain valid, even as a user's behavior, context, and mental-health state evolve. Periodic retraining can incorporate recent data, but it cannot update the model immediately after each new observation and may leave newly arriving users poorly represented. These limitations motivate us to apply online learning to mental-health monitoring, allowing the model to continually adapt as new user data arrive.

2.2 Online Learning

Online learning processes a data stream sequentially. The model predicts with its current state and then updates that state when new information becomes available [35, 40]. This predict-then-update

process allows the model to track a changing data distribution without repeatedly retraining from scratch. Online continual learning further studies how to incorporate new observations while preserving previously learned knowledge. Existing methods address this problem through mechanisms such as adaptive layer freezing, selective sampling, parameter merging, and knowledge-preserving perturbations [26, 32, 45].

Most online and online continual learning methods maintain one model for a single evolving stream. Personalized mental-health monitoring instead contains multiple asynchronous user streams, and each user requires a model state that captures how their sensed behaviors relate to their mental-health state. Per-user online adaptation can update that state as personal data arrive, but a new user initially provides too little evidence for reliable personalization. Therefore, the central challenge is not only adapting over time, but also learning how a low-evidence user can benefit from other users without introducing incompatible updates.

3 Preliminary

3.1 Task Formulation

Let $\mathcal{U}$ be the set of monitored users. User behavioral data and mental-health labels arrive asynchronously as data triplets, forming the full **data stream** $\mathcal{S} = \{(x_t, u_t, y_t)\}_{t=1}^T$. At time t , $x_t \in \mathbb{R}^d$ is the input data of user u_t and contains d behavioral features derived from passive smartphone and wearable sensing, such as sleep, mobility, physical activity, and phone interaction. The binary label y_t represents the user's mental-health state, where $y_t = 1$ denotes the risk state and $y_t = 0$ denotes the non-risk state. In the experiments, the risk state is defined as elevated depressive symptoms derived from the self-reported assessment in each dataset. The model can use only the previous data triplets in $\mathcal{H}_{t-1} = \{(x_i, u_i, y_i) \mid i < t\}$. Based on these concepts, we can define the personalized mental-health monitoring task as follows.

Definition 1 (Personalized mental-health monitoring task). Given (x_t, u_t) and $\mathcal{H}_{t-1}$, the personalized mental-health monitoring task aims to predict the mental-health state of user u at time t , denoted as $\hat{y}_t \in \{0, 1\}$.

3.2 Relation Across Users and Groups

This section studies whether finding similar users is useful for mental-health monitoring. For each user u , the behavioral features x provide implicit signals of the user's mental-health state, while the self-reported label y provides an explicit signal. Therefore, we combine the above two signals to comprehensively estimate user similarity. Formally, for each user u , the relation profile $\mathbf{r}_u$ is computed as

$$\mathbf{r}_u = \mathbb{E}[\tilde{x} \mid y = 1, u] - \mathbb{E}[\tilde{x} \mid y = 0, u], \quad (1)$$

where $\tilde{x} \in \mathbb{R}^d$ is the normalized behavioral data and $\mathbb{E}$ is the mean over user u 's records with the given label. Eq. (1) subtracts the user's average normalized behavior in non-risk records from that in risk records. Each entry of $\mathbf{r}_u \in \mathbb{R}^d$ shows how one behavioral feature differs between the two mental-health states. Two user profiles are compared using cosine similarity.

Taking CES as an example, normal users with at least 20 records and long-history users with at least 160 records are selected. For each user set, the relation profiles are standardized, reduced by PCA,

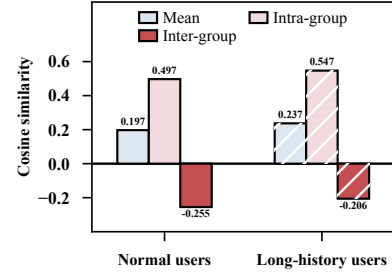


Figure 2: Relations across users in CES. Normal and long-history users have at least 20 and 160 records, respectively.

and divided into three groups using KMeans. Similarity is then calculated from three views. **Mean similarity** compares each user profile with the overall mean profile. **Intra-group similarity** compares user profiles within the same group. **Inter-group similarity** compares the mean profiles of different groups. Figure 2 shows clear groups of users with similar relationships between behavior and mental-health states. This finding supports **OCAP's idea of using similar users to alleviate the low-evidence problem**. It is noted that the above analysis cannot be applied to the prediction phase, as this similarity calculation requires ground-truth labels, which are inaccessible during prediction.

4 Methodology

In this section, we will detail the design of OCAP which consists of basic architecture, gradient-based grouping and group refinement.

4.1 Basic Architecture

OCAP's basic architecture consists of a shared backbone and one lightweight adapter τ_u for each user $u \in \mathcal{U}$. Given user u at time t , the backbone has parameters Θ_{t-1}^b , and the user's adapter has parameters $\Theta_{t-1}^{u_a}$; both are learned from $\mathcal{H}_{t-1}$. Following parameter-efficient adaptation [17], the backbone learns a common feature representation during an initial warm-up period and is then frozen, while τ_u remains trainable and captures user-specific associations between sensed behaviors and mental-health states. The model state also includes the current group set $\mathcal{G}_{t-1}$. Each group $g \in \mathcal{G}_{t-1}$ has a group adapter γ_g with parameters $\Theta_{t-1}^{g_a}$. Moreover, an adapter-routing function π determines which adapter is combined with the backbone for user u at time t . Specifically, $\pi(u, \mathcal{G}_{t-1}) = \tau_u$ when u is not assigned to any group in $\mathcal{G}_{t-1}$, and $\pi(u, \mathcal{G}_{t-1}) = \gamma_g$ when $u \in g$ for some $g \in \mathcal{G}_{t-1}$.

At time t , the data triplet (x_t, u_t, y_t) arrives¹. The model uses its state at $t - 1$ to predict from x_t and then uses y_t to calculate the standard binary cross-entropy loss.

$$\mathcal{L}_t = -y_t \log f(x_t | \Theta_{t-1}^b, \Theta_{t-1}^{\pi(u, \mathcal{G}_{t-1})}) - (1 - y_t) \log [1 - f(x_t | \Theta_{t-1}^b, \Theta_{t-1}^{\pi(u, \mathcal{G}_{t-1})})], \quad (2)$$

where f is the model formed by the shared backbone and the personalized adapter.

¹For better readability, the time subscript t of u_t is omitted below.

Algorithm 1 Gradient-based user grouping.

Input. Users $\mathcal{F}_t$ and their representations $\mathbf{z}_t^u$; minimum group count κ

Output. Updated group set $\mathcal{G}_t$

1. $\mathcal{G}_t \leftarrow \{\{u\} \mid u \in \mathcal{F}_t\}$ and $\mathbf{z}_t^{\{u\}} \leftarrow \mathbf{z}_t^u$
2. **while** $|\mathcal{G}_t| > \kappa$ **do**
3. Find an available pair $a, b \in \mathcal{G}_t$ with the largest $S_t(a, b)$
4. **if** no pair is available or $S_t(a, b) < 0$ **then**
5. **break**
6. **end if**
7. Merge a and b ; update the representation and affected similarities using Eqs. (10) and (9)
8. **end while**
9. Average the state- t user adapters in each $g \in \mathcal{G}_t$ to obtain γ_g
10. Use $\pi(u, \mathcal{G}_t) = \tau_u$ for each user without a valid assignment

Backbone Warm-up. Let $T_{\text{warm-up}}$ denote the number of initial data triplets used to train the shared backbone. Based on $\mathcal{L}_t$, its parameters are updated as

$$\Theta_t^b = \begin{cases} \Theta_{t-1}^b - \eta \nabla_{\Theta_{t-1}^b} \mathcal{L}_t, & t \leq T_{\text{warm-up}}, \\ \Theta_{t-1}^b, & t > T_{\text{warm-up}}. \end{cases} \quad (3)$$

where $\eta > 0$ is the learning rate. After $T_{\text{warm-up}}$ data triplets, the backbone is frozen, and the remaining data are used for personalized adapter learning.

Adapter Learning. Adapter parameters are updated throughout the full data stream. Their updates depend on the user's current group assignment. For a user u without a valid group assignment, the model combines the shared backbone with u 's own adapter τ_u for prediction, and only τ_u is updated.

$$\Theta_t^{\tau u} = \Theta_{t-1}^{\tau u} - \eta \nabla_{\Theta_{t-1}^{\tau u}} \mathcal{L}_t. \quad (4)$$

For a user u assigned to group g , the loss is computed with the group adapter γ_g . The corresponding adapter update direction is

$$\Delta \Theta_t^{\gamma g} = -\eta \nabla_{\Theta_{t-1}^{\gamma g}} \mathcal{L}_t, \quad u \in g. \quad (5)$$

After warm-up, subsequent updates affect only the adapters, leaving the shared representation fixed.

4.2 Gradient-Based Grouping

Task-grouping and clustered federated learning methods show that sharing updates within compatible task or client groups can improve joint learning [11–14, 31]. However, raw behavioral features do not provide a reliable grouping signal for personalized mental-health monitoring. As discussed in Section 3.2, the same behavior may indicate different mental-health states for different users, so similar features do not necessarily imply similar behavior–state relationships. Section 3.2 combines behavioral features with ground-truth labels to study these relationships, but this measure cannot guide prediction because the label of the incoming data is unavailable at that time. Therefore, an online grouping signal must capture user compatibility without the current label. To this end, we propose gradient-based grouping, which represents each user through accumulated model-update directions and groups users that require similar model corrections. Gradients directly describe how

the model needs to change for each user and thus provide a more reliable signal of learning compatibility than raw features [21, 44]. The process first constructs a gradient-based representation for each user and then groups users according to the similarity between their representations.

Gradient-based User Representation. Section 4.1 provides the backbone and selected-adapter gradients for the current user u . To exploit it for user representation, we first map these gradients into the same d_s -dimensional space.

$$\mathbf{g}_t^u = \text{MLP}_{\text{back}}\left(\nabla_{\Theta_{t-1}^b} \mathcal{L}_t\right) + \text{MLP}_{\text{ada}}\left(\nabla_{\Theta_{t-1}^{\pi(u, \mathcal{G}_{t-1})}} \mathcal{L}_t\right), \quad (6)$$

where MLP_{back} maps $\mathbb{R}^{|\Theta_{t-1}^b|}$ to $\mathbb{R}^{d_s}$, and MLP_{ada} maps $\mathbb{R}^{|\Theta_{t-1}^{\pi(u, \mathcal{G}_{t-1})}|}$ to $\mathbb{R}^{d_s}$. The two gradients are projected into the same latent space, and $\mathbf{g}_t^u$ represents the update direction contributed by the current data. This direction is then accumulated in the gradient history of user u .

$$\mathbf{m}_t^u = (1 - \alpha) \mathbf{m}_{t-1}^u + \alpha \frac{\mathbf{g}_t^u}{\|\mathbf{g}_t^u\| + \varepsilon}, \quad (7)$$

where $\varepsilon > 0$ prevents division by zero, $0 < \alpha \leq 1$ controls the contribution of the current direction, and $\mathbf{m}_0^u$ is initialized with the zero vector. Normalizing $\mathbf{g}_t^u$ makes the history depend on the **update direction rather than the gradient magnitude**, while the exponential moving average reduces fluctuations from individual records. Finally, we obtain gradient-based representation of u :

$$\mathbf{z}_t^u = \frac{\mathbf{m}_t^u}{\|\mathbf{m}_t^u\| + \varepsilon}, \quad (8)$$

where $\mathbf{z}_t^u \in \mathbb{R}^{d_s}$ is the final representation of user u . Eq. (8) normalizes the accumulated history so that users are compared by update direction.

Grouping. At time t , $\mathcal{F}_t$ denotes all users observed in the data stream by time t . Each user in $\mathcal{F}_t$ has accumulated at least one model-update direction and thus has a gradient-based representation for grouping. Based on these representations, user groups $\mathcal{G}_t$ can be constructed. Every user $u \in \mathcal{F}_t$ is first treated as a single-user group, giving $\mathcal{G}_t = \{\{u\} \mid u \in \mathcal{F}_t\}$. The representation of the initial group $\{u\}$ is $\mathbf{z}_t^u$. The similarity between every pair of current groups $a, b \in \mathcal{G}_t$ is then calculated as

$$S_t(a, b) = \cos(\mathbf{z}_t^a, \mathbf{z}_t^b), \quad (9)$$

where $\cos(\cdot, \cdot)$ denotes cosine similarity. For two initial single-user groups, Eq. (9) directly compares their user representations.

The two groups with the largest nonnegative similarity are selected and merged into $g = a \cup b$. The representation of the merged group is updated as

$$\mathbf{z}_t^g = \frac{\sum_{u \in g} \mathbf{z}_t^u}{\|\sum_{u \in g} \mathbf{z}_t^u\| + \varepsilon}, \quad (10)$$

where $\mathbf{z}_t^g$ summarizes the update directions of all users in the new group g . After each merge, the new group representation is obtained using Eq. (10), and its similarities with the remaining groups are recalculated using Eq. (9). These updated similarities determine the next two groups to merge.

This process is repeated until κ groups remain or the largest available similarity becomes negative. Once merging stops, the remaining groups form $\mathcal{G}_t$; hence, κ is a minimum target rather

Table 1: Statistical information of datasets.

Dataset	Duration	#Users	#Records	Avg. User Records	Positive Labels (%)
CES [25]	2017–2022	218	35,289	161.9	28.5
GLOBEM [39]	2018–2021	704	8,225	11.7	46.2

than a guarantee on the final group count. For each $g \in \mathcal{G}_t$, its group adapter is constructed as

$$\Theta_t^{y_g} = \frac{1}{|g|} \sum_{u \in g} \Theta_t^{r_u}. \quad (11)$$

where $\Theta_t^{r_u}$ denotes the current adapter parameters of user u , and $|g|$ is the number of users in g . The adapter-routing function then sets $\pi(u, \mathcal{G}_t) = y_g$. Users without a compatible group continue using their own adapters for prediction. Algorithm 1 summarizes this grouping process.

4.3 User Reassignment for Group Refinement

Gradient-based grouping assigns users according to the representations available at an assignment update. These representations summarize the evidence observed so far and change with asynchronous user data. An early assignment may become outdated when additional observations produce a more reliable update direction or reveal that another group adapter better captures the user’s current behavior–label relationship. Keeping such a user in the original group would expose the user to an increasingly mismatched merged adapter and could also reduce the coherence of the adapter shared by the other group members. Therefore, group refinement periodically checks whether each assigned user is better suited to another current group adapter.

For a user u currently assigned to group g , the method evaluates the user’s recent data with every current group adapter. Let $\mathcal{V}_t^u(g')$ denote the mean loss produced by $\gamma_{g'}$. The best available group and its improvement over the current assignment are

$$\begin{aligned} g^+(u) &= \arg \min_{g' \in \mathcal{G}_t} \mathcal{V}_t^u(g'), \\ \Delta_t^{\text{move}}(u) &= \mathcal{V}_t^u(g) - \mathcal{V}_t^u(g^+(u)). \end{aligned} \quad (12)$$

where $g^+(u)$ is the existing group whose adapter gives the lowest recent loss, and $\Delta_t^{\text{move}}(u)$ measures the improvement over the current group adapter. The method considers user u for reassignment only after receiving at least λ_m records from that user, namely $n_t^u \geq \lambda_m$. The reassignment is applied only when $\Delta_t^{\text{move}}(u) \geq \delta_{\text{move}}$. The record threshold λ_m provides enough data for comparing group adapters, while the margin δ_{move} suppresses moves caused by small loss fluctuations. When both conditions hold, u moves to $g^+(u)$, and the adapters of the source and destination groups are recomputed from their current user adapters. The updated $\mathcal{G}_t$ and adapter-routing function π then determine the adapters used for subsequent predictions.

5 Experiments

Experiments are conducted to answer five research questions (RQs).

- **RQ1.** What characterizes mental-health monitoring datasets?

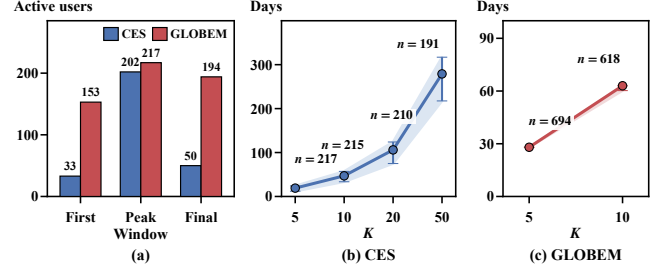


Figure 3: Dataset analysis for CES and GLOBEM. (a) Active users in the earliest window (First), the window with the most active users (Peak), and the latest window (Final). (b–c) Median days required to collect the first K feature–label pairs in CES and GLOBEM; shaded areas and error bars show the interquartile range, and n denotes the number of users reaching each K .

- **RQ2.** How does OCAP perform compared with existing methods?
- **RQ3.** How does OCAP perform when only limited personal evidence is available?
- **RQ4.** How do OCAP components affect performance?
- **RQ5.** How sensitive is OCAP to its key hyperparameters?

5.1 Experimental Setup

Dataset. OCAP is evaluated on two longitudinal passive-sensing datasets for mental-health monitoring, CES [25] and GLOBEM [39]. CES collected daily smartphone and wearable sensing data on mobility, physical activity, sleep, phone interaction, and social context, together with repeated PHQ-4 assessments from 2017 to 2022. GLOBEM collected daily smartphone and Fitbit sensing data on location, screen use, calls, nearby Bluetooth devices, physical activity, and sleep, together with weekly PHQ-4 assessments from 2018 to 2021. For CES, following the PHQ-4 ranges used in prior work [6], scores from 0 to 3 are labeled $y = 0$, and scores from 4 to 12 are labeled $y = 1$. For GLOBEM, the provided binary depression label is used, assigning $y = 0$ to PHQ-4 scores from 0 to 2 and $y = 1$ to scores greater than 2. Table 1 summarizes the dataset information.

Evaluation Metrics. Following prior studies on passive-sensing-based mental-health monitoring [7, 36], we report F1, Accuracy, Recall, and AUC for full-stream performance. F1 is the primary metric because the labels are imbalanced and it jointly captures precision and recall for the mental-health-risk class. Accuracy, Recall, and AUC are used for complementary analysis of overall correctness, risk-class sensitivity, and class discrimination, respectively. To evaluate performance when only limited personal evidence is available, $F1@K$ and $R@K$ are further introduced. Let $\mathcal{U}_K$ be the set of users with at least K records, and let $\mathcal{E}_K^u = \{(y_{u,j}, \hat{y}_{u,j}) \mid 1 \leq j \leq K\}$ contain the labels and prequential predictions for the first K records of user u . F1 is first calculated from each eligible user’s first K predictions and then averaged across users.

$$F1@K = \frac{1}{|\mathcal{U}_K|} \sum_{u \in \mathcal{U}_K} F1(\mathcal{E}_K^u). \quad (13)$$

Table 2: Full-stream performance on CES and GLOBEM. F1 and Recall are computed for the mental-health-risk class, Accuracy is computed over all observations, and AUC measures class discrimination. Online baselines are reported under both global and per-user configurations. Bold and underline denote the best and second-best results for each metric, respectively.

Dataset	Metric	Offline ML						Offline DL				Online (Global)					Online (Per-user)					Ours
		XGB	SVM	LR	RF	LGBM	DT	LSTM	Trans.	TCN	MLP	HBP	KOIL	OLFL	OLI ² DS	OLIFL	HBP	KOIL	OLFL	OLI ² DS	OLIFL	
CES	Recall	0.4500	0.3130	0.4960	0.3070	0.4080	0.4650	0.5790	0.4890	0.3570	0.3860	0.8599	0.9125	0.8784	0.8720	0.8943	0.4955	0.4838	0.6329	0.7436	0.6812	0.7543
	Accuracy	0.5370	0.5702	0.5297	0.5677	0.5377	0.5182	0.5724	0.5700	0.5825	0.5646	0.4254	0.3563	0.3966	0.4168	0.3844	0.7245	0.7575	0.7543	0.7358	<u>0.7945</u>	0.8203
	AUC	0.5240	0.5317	0.5247	0.5287	0.5182	0.5104	0.5753	0.5450	0.5132	0.5099	0.5561	0.5236	0.5416	0.5538	0.5378	0.7457	0.7294	0.8408	0.8104	<u>0.8650</u>	0.8709
	F1	0.4450	0.3750	0.4650	0.3690	0.4210	0.4430	0.4300	0.3870	0.3220	0.3300	0.4602	0.4468	0.4534	0.4600	0.4528	0.5061	0.5320	0.5947	0.6159	<u>0.6538</u>	0.7060
GLOBEM	Recall	0.5150	0.3630	0.5990	0.5030	0.5280	0.5500	0.2630	0.6710	0.4410	0.3810	0.9777	0.9813	0.9753	0.9658	0.9669	0.5551	0.5043	0.7619	0.7703	0.7469	0.8503
	Accuracy	0.5456	0.5503	0.5503	0.5398	0.5450	0.5276	0.5445	0.5404	0.4927	0.4939	0.4707	0.4654	0.4715	0.4744	0.4706	0.5467	0.6900	0.7419	0.7357	<u>0.7409</u>	0.7179
	AUC	0.5436	0.5374	0.5535	0.5373	0.5439	0.5292	0.5251	0.5494	0.4892	0.4862	0.5064	0.5017	0.5070	0.5090	0.5056	0.5685	0.6306	0.8301	0.8011	<u>0.8175</u>	0.7926
	F1	0.5120	0.4280	0.5520	0.5030	0.5180	0.5190	0.3480	0.5750	0.4460	0.4110	0.6306	0.6292	0.6304	0.6294	0.6280	0.5310	0.6006	<u>0.7318</u>	0.7293	0.7271	0.7359

where $F1(\mathcal{E}_K^u)$ is user u 's F1 for the mental-health-risk class. $R@K$ is calculated in the same way by replacing F1 with Recall.

Baseline. We choose two types of baselines. The **offline learning-based** methods include conventional ML models (XGBoost [5], SVM [8], LR [9], random forest [4], LightGBM [19], and decision tree [27]) and DL models (LSTM-Attention [16, 22], Transformer [37], TCN [3], and MLP [28]). The **online learning-based** methods include HBP/ODL [29], KOIL [18], OLI²DS [42], OLFL [40], and OLIFL [43].

Experimental Setting. The data triplets from all users are combined into a single stream and ordered by collection time. At time t , each online method observes (x_t, u_t) and predicts the mental-health state using only the model state learned through time $t - 1$. This prediction is recorded before y_t is revealed and the complete data triplet (x_t, u_t, y_t) is used for one model update. This predict-then-update protocol evaluates every prediction before the model learns from its corresponding label. Each online baseline is evaluated under two settings. In the **global setting**, all users share one model, and every incoming data triplet updates that model. In the **per-user setting**, each user maintains an independent model that is updated only by that user's data triplets.

Implementation Details. For OCAP, a Transformer [37] is used as the backbone, and each user's adapter is implemented using LoRA [17] with rank 4. We set $T_{\text{warm-up}} = 300$ on CES and $T_{\text{warm-up}} = 120$ on GLOBEM. The adapters remain trainable throughout. For gradient-based grouping, $\alpha = 0.20$ on CES and $\alpha = 0.10$ on GLOBEM, while $\kappa = 5$ on both datasets. Group assignments are updated every 100 incoming data triplets on CES and every 120 on GLOBEM. The first assignment targets $\lceil \sqrt{|\mathcal{F}_t|} \rceil$ groups while retaining at least κ . For user reassignment, $\lambda_m = 20$ and $\delta_{\text{move}} = 0.01$ on both datasets.

5.2 Dataset Analysis (RQ1)

To characterize mental-health monitoring datasets, the two datasets are analyzed for asynchronous multi-user streaming and the time required to accumulate personal evidence. Results are shown in Figure 3. Here, asynchronous means that users contribute records at different times and rates, entering and leaving the stream independently. A temporal window is a consecutive period used to divide the data stream. For CES, one window covers three consecutive months. For GLOBEM, one window contains all data collected in one year, and different users participate each year. Figure 3(a)

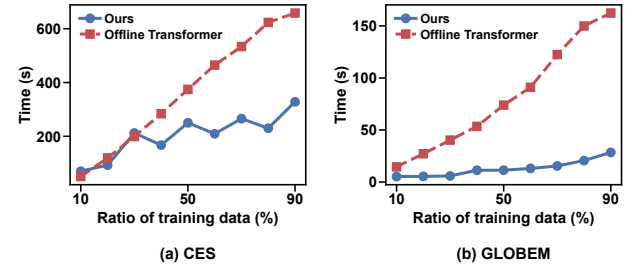


Figure 4: Adaptation cost across stream progress on (a) CES and (b) GLOBEM. At each checkpoint, Offline Transformer retrains on all accumulated data, while Ours processes only the newly arrived 10% interval.

compares the active-user population across three representative windows. First is the earliest window, Peak is the window with the most active users, and Final is the latest window. Figure 3(b) and (c) show the evidence-accumulation results for CES and GLOBEM, respectively.

Figure 3(a) shows that the CES population expands and then contracts, while different users participate in GLOBEM each year. This indicates that both datasets form **asynchronous multi-user streams**. Users provide data at different times, with some users leaving and new users joining over time. Figures 3(b–c) show that collecting more user-specific evidence requires a substantially longer calendar period and that fewer users reach larger K values. This indicates that users may have only limited personal evidence for an extended period, making limited personal evidence a central issue in these datasets. Together, asynchronous participation and slow evidence accumulation require continual adaptation and support before users accumulate sufficient personal evidence.

5.3 Main Results (RQ2)

Table 2 reports the overall performance comparison of OCAP against the baselines. Based on it, we have the following findings.

First, online learning improves both predictive performance and adaptation efficiency for continuously arriving mental-health data. The strongest online methods consistently outperform the offline methods on both datasets. On CES, even the

weakest per-user online baseline exceeds the best offline result by 8.8%, while the strongest online baseline extends this improvement to 40.6%. On GLOBEM, even the weakest global online baseline exceeds the best offline result by 9.2%, and the strongest online baseline achieves a 27.3% improvement. Furthermore, we conduct an **efficiency analysis** to examine whether online adaptation also reduces the cost of processing a growing stream. Figure 4 shows that offline retraining becomes increasingly expensive because it repeatedly processes all accumulated data, whereas OCAP updates only from each newly arrived 10% interval. At 90% stream progress, offline retraining requires 100.6% and 470.2% more time than online adaptation on CES and GLOBEM, respectively. The improvements in both predictive performance and adaptation efficiency demonstrate that online learning is essential for timely mental-health monitoring over continuously arriving data.

Second, user-level personalization is necessary to capture heterogeneous behavior–state relationships and avoid biased common predictions. The per-user settings substantially outperform their global counterparts, showing that a single shared model cannot capture heterogeneous user patterns. On CES, personalization improves OLIFL by 44.4% in F1 and 106.7% in Accuracy. On GLOBEM, it improves OLFL by 16.1% in F1 and 63.7% in AUC. Although the global online methods achieve high Recall, their F1 and Accuracy remain low. Without personalization, these methods tend to learn a common prediction pattern from the imbalanced stream, such as predicting the mental-health-risk class for most users. This behavior covers many positive cases but produces many false positives, so high Recall does not imply strong overall performance.

Third, compatible-user grouping further strengthens personalization by supplementing individual evidence with information from users who have similar update needs. OCAP achieves the best F1 on both datasets. On CES, it also attains the best Accuracy and AUC while maintaining strong Recall, showing consistent performance under the highly imbalanced label distribution. On GLOBEM, its F1 remains the best and its other metrics are comparable to the strongest baselines. These gains come from grouping users by compatible update directions. Averaging their adapters supplements limited personal evidence with relevant information, while reassignment prevents outdated sharing relationships from suppressing personalization. The next section examines this design when only limited personal evidence is available.

5.4 Low-Evidence Performance (RQ3)

Table 3 reports performance over each user’s first K records. **Overall, OCAP outperforms all baselines in every evaluated case.** Its margin over the strongest baseline is consistently larger at $K = 5$ than at $K = 10$, confirming that its advantage is greatest when only a few personal records are available. At this point, a personal adapter learned in isolation is difficult to estimate reliably. Gradient-based grouping identifies adapters that require similar model corrections from their accumulated update directions, and adapter averaging gives each user a broader evidence base without replacing its personal model. Reassignment further updates mismatched sharing relationships as new data arrive. Together, these mechanisms improve predictions under limited personal evidence while limiting interference from incompatible users.

Table 3: Low-evidence performance on CES and GLOBEM. Each metric is computed from every eligible user’s first K predictions and then averaged across users. Bold and underline denote the best and second-best results, respectively.

Method	CES				GLOBEM			
	F1@5	F1@10	R@5	R@10	F1@5	F1@10	R@5	R@10
HBP/ODL	0.1300	0.1409	0.1882	0.2178	0.2772	0.3129	0.3512	0.4009
KOIL	0.2060	0.1861	0.2277	0.2016	0.3490	0.3031	0.3622	0.2932
OLFL	0.2162	0.2100	0.2773	0.2811	0.3829	0.4252	0.4280	0.4538
OLI ² DS	0.2322	<u>0.2427</u>	0.2888	<u>0.3015</u>	0.4060	<u>0.4585</u>	<u>0.4334</u>	<u>0.4769</u>
OLIFL	0.2328	0.2267	<u>0.2930</u>	0.2951	0.3911	0.4222	0.4241	0.4427
OCAP	<u>0.2713</u>	<u>0.2632</u>	<u>0.3848</u>	<u>0.3932</u>	<u>0.4380</u>	<u>0.4697</u>	<u>0.5180</u>	<u>0.5445</u>

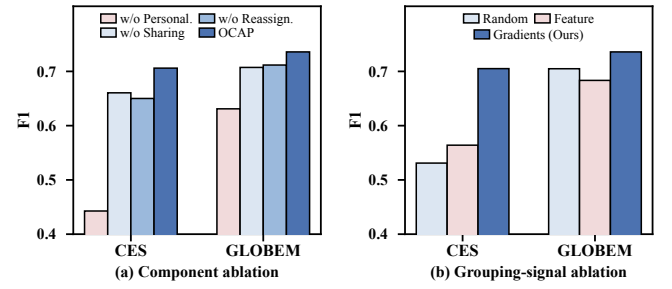


Figure 5: Full-stream F1 ablations on CES and GLOBEM. (a) Component ablation. (b) Grouping-signal ablation.

5.5 Ablation Study (RQ4)

We first construct three variants to evaluate the contribution of each component in OCAP. **w/o Personalization** removes the user adapters and makes predictions with the shared backbone alone. **w/o Sharing** disables user grouping while retaining an independent adapter for each user. **w/o Reassignment** retains each user’s initial group assignment for all later data pairs. In Section 4.2, we propose gradient-based grouping to identify users with compatible update directions. To evaluate this grouping signal, we construct two additional variants while keeping the remaining components unchanged. **Random** assigns users to groups randomly, whereas **Feature** groups users according to their standardized behavioral features. The results are shown in Figure 5.

Effect of Components. Figure 5(a) shows that OCAP consistently performs best on both datasets. Removing personalization causes the largest degradation, especially on CES, confirming that a shared backbone alone cannot capture heterogeneous user patterns. Removing group sharing also reduces F1 because each user must then learn only from its own limited data. Performance further declines without user reassignment, showing that fixed groups become less suitable as newly arriving data change users’ update directions. These degradations show that the three components provide complementary benefits.

Effect of the Grouping Signal. Figure 5(b) shows that random and feature-based grouping vary across the two datasets, and feature-based grouping is even weaker than random assignment on GLOBEM. This instability indicates that similar observed behaviors

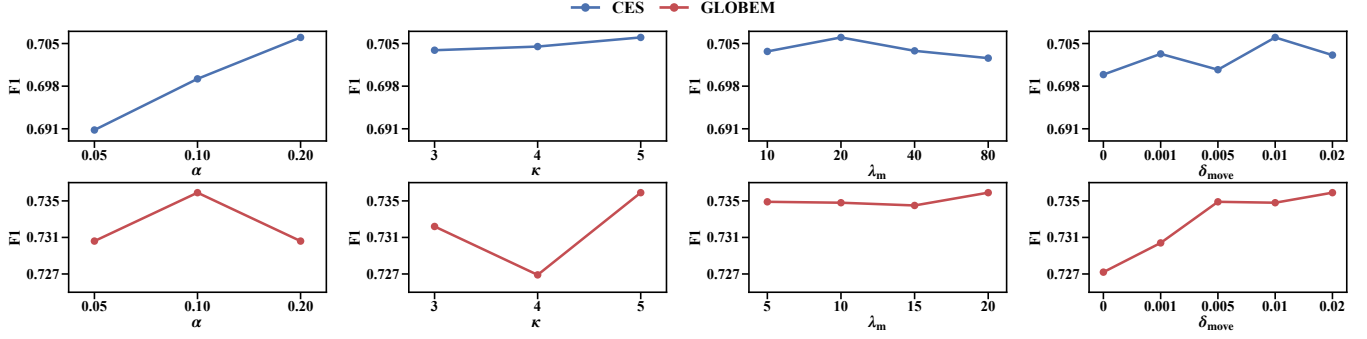


Figure 6: Full-stream F1 sensitivity to four hyperparameters on CES and GLOBEM. The λ_m plots fix $\delta_{move} = 0.01$; the δ_{move} plots fix $\lambda_m = 20$ on CES and $\lambda_m = 10$ on GLOBEM.

do not consistently imply compatible mental-health prediction patterns. In contrast, gradient-based grouping achieves the highest F1 on both datasets because update directions directly describe how each user’s adapter needs to learn. Therefore, they reliably identify users that can share.

5.6 Hyperparameter Analysis (RQ5)

Figure 6 shows that α and δ_{move} are the more sensitive hyperparameters, λ_m is stable on both datasets, and the sensitivity to κ depends on the dataset. For α , CES improves as more weight is placed on the current update direction, whereas GLOBEM peaks at an intermediate value. CES has much longer user histories than GLOBEM, so its update directions are better supported and a larger α can follow recent changes; the sparser GLOBEM histories require stronger smoothing to reduce the effect of noisy updates. The movement margin δ_{move} also has a visible effect because a small value accepts marginal and potentially unstable reassignments, while an overly large value can block useful changes. Performance becomes stable on GLOBEM once a moderate margin is applied and peaks at 0.01 on CES. In contrast, changing λ_m produces only minor variation, indicating that reassignment is not sensitive to the exact evidence threshold once a basic amount of user history is available. Finally, κ has little effect on CES but causes larger fluctuations on GLOBEM because sparse user histories make its group construction more sensitive to the selected number of groups.

6 Conclusion

This paper studies personalized mental-health monitoring from asynchronous passive-sensing data streams. Our analysis reveals that fixed offline models cannot adapt promptly to evolving associations between sensed behaviors and mental-health states, while online personalization is constrained by the slow accumulation of personal data. Sharing information across users can supplement this limited evidence, but differences in users’ behavior–state relationships and their evolution over time make fixed sharing relationships unreliable. These challenges motivate the central idea of learning personal models continually, sharing information only among users with compatible learning needs, and revising their sharing relationships as new evidence arrives. Following this idea, OCAP uses a shared backbone to learn common patterns and continually updated

personal adapters to capture user-specific patterns. It represents users through their accumulated model-update directions, groups users with compatible directions, averages their adapters for prediction, and reassigns users when their current groups no longer fit their recent data. Experiments on multiple real-world datasets validate the effectiveness of OCAP.

7 Limitations and Ethical Considerations

Limitations. OCAP is evaluated on CES and GLOBEM, which primarily contain university-student populations. Its generalizability to other populations, sensing devices, and clinical settings remains to be established. The online updates assume that self-reported labels eventually become available, whereas deployment may involve longer feedback delays or missing labels. The experiments also reduce mental-health risk to a binary prediction task and do not evaluate clinical outcomes. Further work should examine the frozen backbone, gradient-based grouping, and reassignment across broader sensing modalities, label schedules, and populations.

Ethical considerations. This study uses the de-identified CES and GLOBEM datasets, whose original collection protocols received institutional ethical approval and obtained participant consent [25, 39]. Passive-sensing data and learned user representations may reveal sensitive behavior, so future use requires informed consent, data minimization, secure access, and privacy auditing. Differences in population coverage, device access, missing data, and feedback frequency may also bias predictions and adapter sharing, so deployment should evaluate subgroup false-positive and false-negative rates. OCAP is a decision-support tool and should not be used for automatic diagnosis, punitive decisions, employment or insurance screening, or covert surveillance. Errors may delay support or trigger unnecessary intervention, so human oversight is essential throughout deployment.

8 Generative AI Usage

Generative AI tools assisted with language editing, code refinement, and figure presentation. The authors verified every suggestion and remain responsible for this work.

References

- [1] Daniel A. Adler, Fei Wang, David C. Mohr, and Tanzeem Choudhury. 2022. Machine learning for passive mental health symptom prediction: Generalization across different longitudinal mobile sensing studies. *PLOS ONE* 17, 4 (2022), e0266516.
- [2] C. Akre et al. 2024. Advancing digital sensing in mental health research. *npj Digital Medicine* 7 (2024), 362.
- [3] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. 2018. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv preprint arXiv:1803.01271* (2018). doi:10.48550/arXiv.1803.01271
- [4] Leo Breiman. 2001. Random forests. *Machine Learning* 45, 1 (2001), 5–32.
- [5] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 785–794.
- [6] Meghna Roy Chowdhury, Wei Xuan, Shreyas Sen, Yixue Zhao, and Yi Ding. 2025. Predicting and understanding college student mental health with interpretable machine learning. In *Proceedings of the ACM/IEEE International Conference on Connected Health: Applications, Systems and Engineering Technologies*. 1–12.
- [7] Filippo Corponi, Bryan M. Li, Gerard Anmella, Ariadna Mas, Isabella Pacchiarotti, Marc Valenti, Iria Grande, Antoni Benabarre, Marina Garriga, Eduard Vieta, Stephen M. Lawrie, Heather C. Whalley, Diego Hidalgo-Mazzei, and Antonio Vergari. 2024. Automated mood disorder symptoms monitoring from multivariate time-series sensory data: Getting the full picture beyond a single number. *Translational Psychiatry* 14, 1 (2024), 161.
- [8] Corinna Cortes and Vladimir Vapnik. 1995. Support-vector networks. *Machine Learning* 20, 3 (1995), 273–297.
- [9] D. R. Cox. 1958. The regression analysis of binary sequences. *Journal of the Royal Statistical Society: Series B (Methodological)* 20, 2 (1958), 215–232.
- [10] A. Doherty, S. Bucci, A. Kenny, R. Kotov, G. Lipinska, L. Ospina-Pinillos, and K. Schultebrucks. 2025. Passive sensing at scale to advance the understanding of mental health. *The Lancet Digital Health* 7, 3 (2025), e172–e174.
- [11] Christopher Fifty, Ehsan Amid, Zhe Zhao, Tianhe Yu, Rohan Anil, and Chelsea Finn. 2021. Efficiently identifying task groupings for multi-task learning. In *Proceedings of the 35th International Conference on Neural Information Processing Systems*, Vol. 34. 27503–27516.
- [12] Yuan Gao, Shuguo Jiang, Moran Li, Jin-Gang Yu, and Gui-Song Xia. 2024. DMTG: One-shot differentiable multi-task grouping. In *Proceedings of the 41st International Conference on Machine Learning*, Vol. 235. 14747–14762.
- [13] Avishek Ghosh, Jichan Chung, Dong Yin, and Kannan Ramchandran. 2020. An efficient framework for clustered federated learning. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, Vol. 33. 19586–19597.
- [14] Yongxin Guo, Xiaoying Tang, and Tao Lin. 2025. Enhancing clustered federated learning: Integration of strategies and improved methodologies. In *Proceedings of the 13th International Conference on Learning Representations*.
- [15] Xiangnan He, Kuan Deng, Xiang Wang, Yan Li, Yongdong Zhang, and Meng Wang. 2020. LightGCN: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*. 639–648.
- [16] Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long short-term memory. *Neural Computation* 9, 8 (1997), 1735–1780.
- [17] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *Proceedings of the 10th International Conference on Learning Representations*.
- [18] Junjie Hu, Haiqin Yang, Irwin King, Michael R. Lyu, and Anthony Man-Cho So. 2015. Kernelized online imbalanced learning with fixed budgets. In *Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence*, Vol. 29. 2666–2672.
- [19] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. LightGBM: A highly efficient gradient boosting decision tree. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, Vol. 30. 3146–3154.
- [20] Robert A Lewis, Asma Ghandeharioun, Szymon Fedor, Paola Pedrelli, Rosalind Picard, and David Mischoulon. 2023. Mixed effects random forests for personalised predictions of clinical depression severity. *arXiv preprint arXiv:2301.09815* (2023).
- [21] Dongyue Li, Aneesh Sharma, and Hongyang R. Zhang. 2024. Scalable multitask learning using gradient-based estimation of task affinity. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 1542–1553.
- [22] Thang Luong, Hieu Pham, and Christopher D. Manning. 2015. Effective approaches to attention-based neural machine translation. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*. 1412–1421.
- [23] David C. Mohr, Mi Zhang, and Stephen M. Schueller. 2017. Personal sensing: Understanding mental health using ubiquitous sensors and machine learning. *Annual Review of Clinical Psychology* 13 (2017), 23–47.
- [24] Tahsin Mullick, Sam Shaaban, Ana Radovic, and Afsaneh Doryab. 2024. Framework for ranking machine learning predictions of limited, multimodal, and longitudinal behavioral passive sensing data: Combining user-agnostic and personalized modeling. *JMIR AI* 3 (2024), e47805.
- [25] Subigya Nepal et al. 2024. Capturing the college experience: A four-year mobile sensing study of mental health, resilience and behavior of college students during the pandemic. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 8, 1, Article 38 (2024). doi:10.1145/3643501
- [26] Benliu Qiu, Heqian Qiu, Lanxiao Wang, Taijin Zhao, Yu Dai, Lili Pan, and Hongliang Li. 2026. Parameter merging with gradient-guided supermasks in online continual learning. In *Proceedings of the 40th AAAI Conference on Artificial Intelligence*, Vol. 40. 24991–24999.
- [27] J. R. Quinlan. 1986. Induction of decision trees. *Machine Learning* 1, 1 (1986), 81–106.
- [28] David E. Rumelhart, Geoffrey E. Hinton, and Ronald J. Williams. 1986. Learning representations by back-propagating errors. *Nature* 323 (1986), 533–536.
- [29] Doyen Sahoo, Quang Pham, Jing Lu, and Steven C. H. Hoi. 2018. Online deep learning: Learning deep neural networks on the fly. In *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence*. 2660–2666.
- [30] Alireza Sameh, Mehrdad Rostami, Mourad Oussalah, Raija Korpelainen, et al. 2025. Digital phenotypes and digital biomarkers for health and diseases: A systematic review of machine learning approaches utilizing passive non-invasive signals collected via wearable devices and smartphones. *Artificial Intelligence Review* 58, Article 66 (2025). doi:10.1007/s10462-024-11009-5
- [31] Felix Sattler, Klaus-Robert Müller, and Wojciech Samek. 2021. Clustered federated learning: Model-agnostic distributed multi-task optimization under privacy constraints. *IEEE Transactions on Neural Networks and Learning Systems* 32, 8 (2021), 3710–3722.
- [32] Minhyuk Seo, Hyunseo Koh, and Jonghyun Choi. 2025. Budgeted online continual learning by adaptive layer freezing and frequency-based sampling. In *Proceedings of the 13th International Conference on Learning Representations*.
- [33] Shiyang Shen, Wenhao Qi, Jianwen Zeng, Shiyu Li, Xiaoyu Liu, Xiaoyan Zhu, Chao Dong, Bin Wang, Ying Shi, Jia Yao, Bing Wang, Xiaoting Lou, Shanshan Gu, Ping Li, Jian Wang, Guozhen Jiang, and Shousong Cao. 2025. Passive sensing for mental health monitoring using machine learning with wearables and smartphones: Scoping review. *Journal of Medical Internet Research* 27 (2025), e77066.
- [34] Caitlin A. Stamatis, Jonah Meyerhoff, Yixuan Meng, Zhi Chong Chris Lin, Young Min Cho, Tony Liu, Chris J. Karr, Tingting Liu, Brenda L. Curtis, Lyle H. Ungar, and David C. Mohr. 2024. Differential temporal utility of passively sensed smartphone features for depression and anxiety symptom prediction: A longitudinal cohort study. *npj Mental Health Research* 3 (2024), 1.
- [35] Muhammad Sulaiman, Mina Farmanbar, Shingo Kagami, Ahmed Nabil Belbachir, et al. 2025. Online deep learning’s role in conquering the challenges of streaming data: A survey. *Knowledge and Information Systems* 67 (2025), 3159–3203.
- [36] Adela C. Timmons, Abdullah Aman Tutul, Kleantes Avramidis, Jacqueline B. Duong, Kayla E. Carta, Sierra N. Walters, Grace A. Jumonville, Alyssa S. Carrasco, Gabrielle F. Freitag, Daniela N. Romero, Matthew W. Ahle, Jonathan S. Comer, Shrikanth S. Narayanan, Ishita P. Khurd, and Theodora Chaspari. 2025. Developing personalized algorithms for sensing mental health symptoms in daily life. *npj Mental Health Research* 4, 1, Article 34 (2025). doi:10.1038/s44184-025-00147-5
- [37] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, Vol. 30. 5998–6008.
- [38] Xuhai Xu, Prerna Chikeral, Janine M Dutcher, Yasaman S Sefidgar, Woosuk Seo, Michael J Tumminia, Daniella K Villalba, Sheldon Cohen, Kasey G Creswell, J David Creswell, et al. 2021. Leveraging collaborative filtering for personalized behavior modeling: A case study of depression detection among college students. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 5, 1 (2021), 1–27.
- [39] Xuhai Xu, Xin Liu, Han Zhang, Weichen Wang, Subigya Nepal, Yasaman Sefidgar, Woosuk Seo, Kevin S. Kuehn, Jeremy F. Huckins, Margaret E. Morris, Paula S. Nurius, Eve A. Riskin, Shwetak Patel, Tim Althoff, Andrew Campbell, Anind K. Dey, and Jennifer Mankoff. 2022. GLOBEM: Cross-dataset generalization of longitudinal human behavior modeling. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 4, Article 190 (2022), 34 pages.
- [40] Huigui Yan, Dianlong You, Zhen Chen, and Limin Shen. 2026. Online learning for fickle data streams without labels. *IEEE Transactions on Knowledge and Data Engineering* 38 (2026), 3022–3036.
- [41] Mingqiang Yang, Edith C.-H. Ngai, Xiping Hu, Bin Hu, Jiangchuan Liu, Erol Gelenbe, and Victor C. M. Leung. 2024. Digital phenotyping and feature extraction on smartphone data for depression detection. *Proc. IEEE* 112, 12 (2024), 1773–1798.
- [42] Dianlong You, Jiawei Xiao, Yang Wang, Huigui Yan, Di Wu, Zhen Chen, Limin Shen, and Xindong Wu. 2023. Online learning from incomplete and imbalanced data streams. *IEEE Transactions on Knowledge and Data Engineering* 35, 10 (2023), 10650–10665.
- [43] Dianlong You, Huigui Yan, Jiawei Xiao, Zhen Chen, Di Wu, Limin Shen, and Xindong Wu. 2024. Online learning for data streams with incomplete features

1045	and labels. <i>IEEE Transactions on Knowledge and Data Engineering</i> 36, 9 (2024), 4820–4834.	<i>Signal Processing</i> . IEEE, 1–5.	1103
1046		[45] Dulan Zhou, Zijian Gao, and Kele Xu. 2026. Perturbing to preserve: Defending fragile knowledge in online continual learning. In <i>Proceedings of the 40th AAAI Conference on Artificial Intelligence</i> , Vol. 40. 28937–28945.	1104
1047	[44] Yuxin Zhang, Haoyu Chen, Zheng Lin, Zhe Chen, and Jin Zhao. 2025. LCFed: An efficient clustered federated learning framework for heterogeneous data. In <i>Proceedings of the 2025 IEEE International Conference on Acoustics, Speech and</i>		1105
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